

Modular Winding Arrangement Design to Suppress Short-Circuit Current and Magnetic Coupling in PM Machines for Flywheel Battery

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Due to the advantages of long life-period and environmental friendliness, the flywheel battery is widely concerned. In this paper, a modular winding arrangement is proposed to suppress the short-circuit current (SCC) and magnetic coupling of permanent magnet (PM) machines for flywheel battery. Firstly, the design principle of the modular arrangement is elaborated. Then, the magnetomotive force (MMF) harmonic components in the conventional and modular arrangement are derived through winding spatial distribution. Afterwards, the inductance expression is obtained according to winding function theory. Furthermore, the inversely proportional relationship between self-inductance and SCC are deduced. Then, the finite element method (FEM) is adopted, which verifies that the modular winding arrangement has the advantage of SCC and magnetic coupling suppression in PM machines for flywheel battery. Finally, a five-phase PM machine and the proposed modular arrangement machine, and the experiments are carried out for theoretical verification.

Keywords : modular winding arrangement, PM machines, flywheel battery, SCC, magnetic coupling

1. Introduction

With the continuous enhancement of environmental awareness, the flywheel battery is attracting increasing attention [1, 2]. PM machines have the merits of high efficiency, high-power density and simple structure, which makes it one of core driving components of flywheel battery [3, 4]. However, the high SCC amplitude and magnetic coupling degree in the PM machines will lead to overheating, burned coils and PM irreversible demagnetization [5], which restricts the safety operation of flywheel battery.

As is known to all, the SCC can be effectively restrained with high self-inductance. The relevant assumptions of mathematical analysis model for SCC have been proposed in [6]. A closed-slot 6-slot/4-pole PM fault-tolerant machine was presented in [7]. Compared open-slot machine, the presented machine can enhance self-inductance, improve the sine degree of back electromotive force (EMF) and suppress cogging torque. However, this structure can lead to serious leakage magnetic

flux and local saturation in tooth tips. In [8], the fault-tolerant machines with auxiliary teeth added in each slot were designed to suppress the SCC. A new topology with stator teeth shift was proposed in [9], this structure can effectively improve fault-tolerant capability. But this structure is quite complex, which increases the manufacturing cost. The same drawback also exists in [10]. In addition, the fault-tolerant improvement with optimal phase shift angles in various slot/pole combinations were proposed in [5, 11, 12]. For a 24-slot/10-pole PM machine, the winding arrangement with 15° phase shift angle has the smallest SCC amplitude due to its highest self-inductance among the four phase shift angles (0°, 15°, 30° and 60°) [11]. However, the opposite conclusion was obtained in [12], the winding arrangement with 15° phase shift angle has the largest SCC amplitude in a 24-slot/22-pole PM machine. Besides, in a 48-slot/22-pole PM machine, an unconventional phase shift angle (7.5°) was verified to be the best choice for SCC suppression compared to other angles (15° and 30°) [5]. These winding configurations focus on the SCC suppression, but there is little investigation on magnetic coupling suppression.

In order to suppress SCC amplitude and magnetic coupling at the same time in the PM machines for

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flywheel battery, a modular winding arrangement is proposed in this paper. In Section 2, the design principle of this modular winding arrangement is described in detail. The theoretical analysis including MMF harmonics components, inductance and SCC are investigated in Section 3. In Section 4, the finite element method is used to illustrate the correctness of the theoretical analysis. Finally, the conclusions are drawn in Section 5.

2. Modular Winding Arrangement

In this section, a modular winding arrangement with the purpose of suppressing the SCC and improving magnetic isolation capability of PM machine for flywheel battery is designed, and its principle has been illustrated in detail.

For conventional winding arrangement, as shown in Fig. 1(a), it can be seen that each phase winding is uniformly distributed along the whole circumference. However, each phase winding in modular arrangement is located in diagonally, as shown in Fig. 1(b). The corresponding phasor diagrams are shown in Fig. 2. And the main parameters of the two PM machines are listed in Table 1.

In addition, the isolation teeth have been added between phases to further enhance magnetic isolation capability.

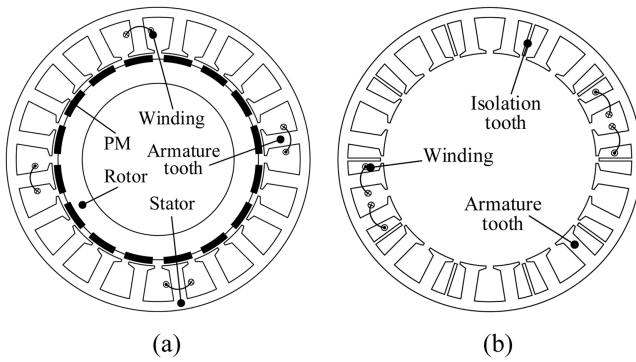


Fig. 1. Cross section of PM machines with different structures: (a) Conventional, (b) Modular.

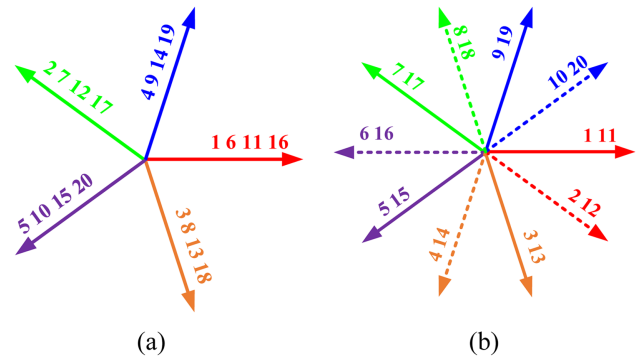
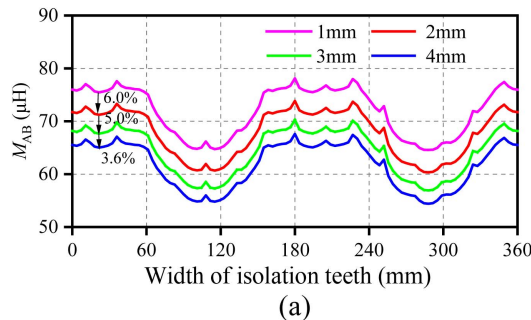


Fig. 2. (Color online) Phasor diagrams for different structures: (a) Conventional, (b) Modular.

Table 1. Main parameters of 20-slot/16-pole PM machines with different winding arrangement.

Quantity	Conventional (Unit)	Modular (Unit)
Slot number	20	
Pole pairs	8	
Stator outer diameter	200 mm	
Stator inner diameter	140 mm	
Stack length	50 mm	
Air-gap length	1.5 mm	
Thickness of PM	4 mm	
PM pole arc coefficient	0.7	
Magnet material	N42UH	
Iron core material	B20AT1500	
Length of isolation teeth	–	22 mm
Width of isolation teeth	–	2 mm

The width and length of the isolation teeth can be determined by the impact on the mutual inductance. Fig. 3 compares the mutual inductance with different width of the isolation teeth. When the width of isolation teeth is 2 mm, the proportion of mutual inductance between phases A and B (M_{AB}) reduction is the largest. With the increase

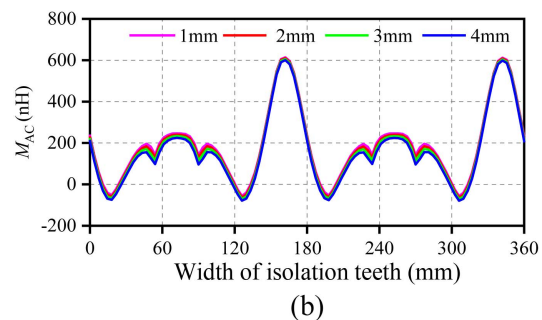


Fig. 3. (Color online) Comparison of mutual inductance with different isolation teeth width: (a) M_{AB} , (b) M_{AC} .

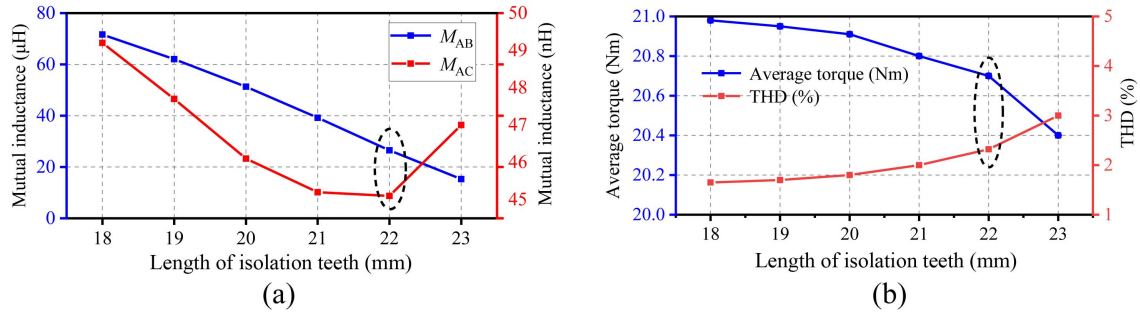


Fig. 4. (Color online) Comparison of machine performance with different isolation teeth length: (a) Mutual inductance, (b) Average torque and THD of back-EMF.

of width, the mutual inductance between phases A and C (M_{AC}) remains almost unchanged. Therefore, the final width of the isolation teeth is 2 mm, and its length will be determined based on this size.

The influence of different isolation teeth length on machine performance, including mutual inductance, average torque and THD of back-EMF, are compared in Fig. 4. When the length of isolation teeth exceeds 22 mm, the value of M_{AC} rebounds. At the same time, the average torque decreases sharply and the THD of back-EMF increases significantly. Finally, the length of isolation teeth is selected as 22 mm. In addition, it should be noted that the THD means the total harmonics distortion, which can be used to evaluate the sine degree of waveforms and expressed as

$$\text{THD} = \sqrt{\frac{\sum_{\lambda \neq 1} E_{\lambda}^2}{E_1^2}} \times 100\% \quad (1)$$

where E_1 is the back-EMF fundamental amplitude, λ and E_{λ} are the harmonic order and amplitude of back-EMF, respectively.

3. Evaluation Principle

3.1. MMF Harmonic

The electromagnetic Taking phase A as an example, its winding function can be derived through turns function, and expressed as [13]

$$N_A(\theta) = n_A(\theta) - \text{ave}[n_A(\theta)] \quad (2)$$

where $N_A(\theta)$ is the winding function of phase A, θ is the angular position in stator reference from phase A axis, $n_A(\theta)$ and $\text{ave}[n_A(\theta)]$ is the turns function and its average, respectively. The winding functions of the two arrangements are shown in Fig. 5. The winding function in conventional arrangement takes $\pi/2$ as a period, while that

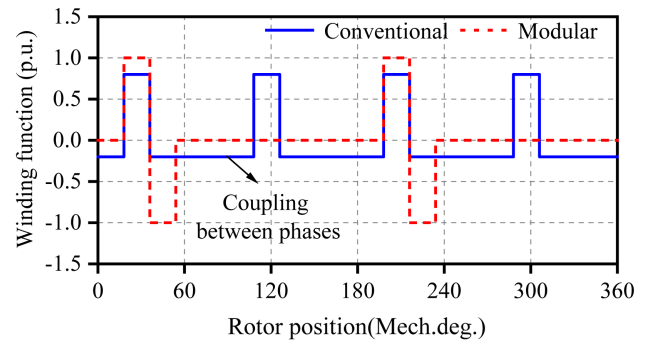


Fig. 5. (Color online) Winding function of two arrangements.

period in modular arrangement is π . To simplify calculation, the Fourier expansion expression of conventional winding function within one period can be written as

$$N_A(\theta) = \sum_{k \neq 5} \sin\left(\frac{k\pi}{5}\right) \cos\left(k\theta - \frac{3k\pi}{5}\right) \quad (3)$$

where k is the harmonic order. Similarly, the winding functions of other four phases can be derived as

$$\begin{cases} N_B(\theta) = \cos\left[k\left(\theta + \frac{4\pi}{5}\right) - \frac{3k\pi}{5}\right] \\ N_C(\theta) = \cos\left[k\left(\theta - \frac{2\pi}{5}\right) - \frac{3k\pi}{5}\right] \\ N_D(\theta) = \cos\left[k\left(\theta + \frac{2\pi}{5}\right) - \frac{3k\pi}{5}\right] \\ N_E(\theta) = \cos\left[k\left(\theta - \frac{4\pi}{5}\right) - \frac{3k\pi}{5}\right] \end{cases} \quad (4)$$

Assuming the phases A, B, C, D and E windings are symmetrical and all have the identical turns number. The combined MMF distribution F of the five-phase windings can be written as

$$F = N_A i_A + N_B i_B + N_C i_C + N_D i_D + N_E i_E \quad (5)$$

where N_A , N_B , N_C , N_D and N_E are the winding function of

the corresponding phase, i_A, i_B, i_C, i_D and i_E are the phase current of each phase, and can be written as

$$\begin{cases} i_A = I_N \cos(\omega t) \\ i_B = I_N \cos(\omega t - 2\pi/5) \\ i_C = I_N \cos(\omega t - 4\pi/5) \\ i_D = I_N \cos(\omega t - 6\pi/5) \\ i_E = I_N \cos(\omega t - 8\pi/5) \end{cases} \quad (6)$$

where I_N is the amplitude of phase current, ω is the angular speed, t is time. F in one period can be finally obtained through substituting (3), (4) and (6) into (5), and written as

$$\begin{aligned} F = & \frac{1}{2} \cos(k\theta - \frac{3k\pi}{5} + \omega t) [1 + 2 \cos(\frac{4k\pi - 2\pi}{5}) + 2 \cos(\frac{2k\pi + 4\pi}{5})] \\ & - \frac{1}{2} \cos(k\theta - \frac{3k\pi}{5} - \omega t) [1 + 2 \cos(\frac{4k\pi + 2\pi}{5}) + 2 \cos(\frac{2k\pi - 4\pi}{5})] \end{aligned} \quad (7)$$

where k represents integer. The conventional winding function is composed of four periods, so the harmonic components in this arrangement can be concluded as $4(5k \pm 2)$. Similarly, it can be inferred that the harmonic components in the modular arrangement is $2(5k \pm 1)$.

3.2. Self-inductance

The self-inductance expression can be derived through winding function theory and written as [14]

$$L_{AA}(\theta) = \frac{\mu_0 l_e r_a}{g} \int_0^{2\pi} N_A^2(\theta) d\theta \quad (8)$$

where L_{AA} is the self-inductance of phase A, μ_0 is the permeability of vacuum, l_e is the effective length of machine and r_a is the radius of air-gap. g is the thickness of air-gap. The expression can be further transformed as

$$L_{AA}(\theta) = L_0 + \sum_j L_{jk} \cos(jk\theta + \beta_{jk}) \quad (9)$$

where j is any non-zero even number, L_{jk} and β_{jk} represent the amplitude value and initial phase of $(jk)^{\text{th}}$ self-inductance harmonic, L_0 is the DC component of the self-inductance, and can be written as

$$L_0 = \frac{\pi \mu_0 r_a l_e}{g} \sum_k \left(\frac{2N_c k_{wk}}{\pi k} \right)^2 \quad (10)$$

where N_c is the turn number of each phase, k_{wk} is the winding factor of the k^{th} MMF harmonic. It can be seen from (10) that the content of the MMF harmonic has a great contribution to self-inductance. According to the analysis in the previous section, the MMF harmonics in

modular arrangement are more abundant than that of in conventional arrangement. Therefore, compared to the conventional arrangement, the modular arrangement has more potential to enhance self-inductance.

3.3. SCC

When a short-circuit fault occurs in one phase (assumed to phase A), a large SCC amplitude will lead to coil insulation damage, overheating and PM irreversible demagnetization, which is not conducive to the safe operation of PM machine for flywheel battery. Thus, the factors affecting the SCC will be analyzed in detail. Firstly, make the following assumptions [6]: the winding in phase A is short-circuited terminally, the phase resistance and rotating speed during short-circuit period keep unchanged, the mutual inductance between phases can be dismissed.

Fig. 6 shows the equivalent circuit diagram with phase A short-circuit fault, and the differential equations can be obtained according to the diagram and written as

$$\begin{cases} \psi_A = L_{AA} i_s + N_c \psi_{PM(0)} \cos \varphi \\ U_\Delta = R_A i_s + \frac{\partial \psi_A}{\partial t} = 0 \end{cases} \quad (11)$$

where ψ_A is the phase A winding flux linkage, i_s is the SCC, $\psi_{PM(0)}$ is the initial value of PM linkage, φ is the electrical angle between the d-axis and the axis of the short-circuited phase winding, U_Δ is the terminal voltage, R_A is the phase resistance. The above equation can be transferred through Laplace transformation, and the SCC expression in time domain can be expressed as

$$\begin{aligned} i_s(t) = & N_c \psi_{PM(0)} \omega \frac{R_A \sin(\omega t + \varphi_0) - \omega L_{AA} \cos(\omega t + \varphi_0)}{R_A^2 + (\omega L_{AA})^2} \\ & + \left\{ I_N \cos \varphi_0 + \frac{N_c \psi_{PM(0)} \omega (\omega L_{AA} \cos \varphi_0 - R_A \sin \varphi_0)}{R_A^2 + (\omega L_{AA})^2} \right\} e^{-\frac{R_A}{L_{AA}} t} \end{aligned} \quad (12)$$

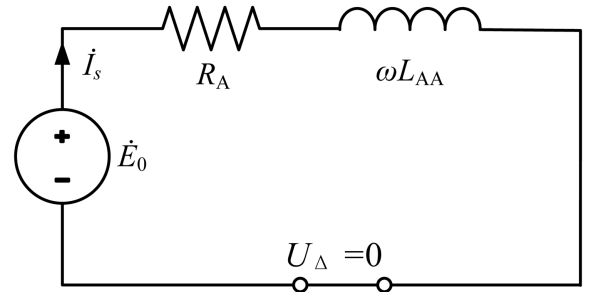


Fig. 6. The equivalent circuit diagram with phase-A short-circuit fault.

where φ_0 is the initial value of φ . As can be known in (12), the SCC expression can be divided into transient and steady part, and the latter can be represented as

$$i_{ss}(t) = \frac{N_c \psi_{PM(0)}}{L_{AA}} \cos(\omega t + \sigma) \quad (13)$$

where σ can be written as

$$\sigma = \arctan\left(-\frac{\omega L_{AA}}{R_A}\right) \quad (14)$$

Furthermore, in case of ωL_{AA} is much greater than R_A , the phase resistance R_A can be ignored and the simplified SCC expression can be obtained as written as

$$i_{ss}(t) = \frac{N_c \psi_{PM(0)}}{L_{AA}} \cos(\omega t + \sigma) \quad (15)$$

It is evident from (15) that the effective method to suppress SCC amplitude is improving the self-inductance.

4. Comparison and Verification

In order to verify the rationality of the above theoretical analysis and the improvement of fault-tolerant capability in the PM machines for flywheel battery. The electromagnetic performances are compared, including MMF harmonic, inductance, SCC amplitude and torque performance.

4.1. MMF Harmonic

Fig. 7 compares the MMF harmonic spectrums of two winding arrangements, the modular arrangement has more harmonic components than that in conventional one. Specifically, the 2nd-, 8th-, 12th- and 18th-, etc. order harmonics are included in modular arrangement, which satisfies the $2(5k \pm 1)$. In addition, 8th- and 12th-, etc. order harmonics exist in conventional arrangement, which meets the $4(5k \pm 2)$. Therefore, the result indicates that the derivation of (7) is reasonable.

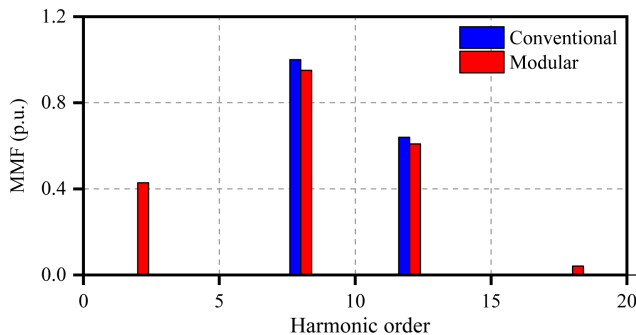


Fig. 7. (Color online) MMF harmonic comparison.

4.2. Self-inductance and SCC

The self-inductance and SCC of two winding arrangements are compared in this section. Fig. 8 shows the self-inductance waveforms of different arrangements, the average value of conventional one is 1.34 mH, while that in modular one is 2.1 mH, with an approximate 60 % increase. It illustrates that the modular arrangement can effectively enhance the self-inductance, thereby suppressing the SCC amplitude, as shown in Fig. 9.

It can be seen that the steady SCC amplitudes in conventional and modular arrangements are 68.3 A and 43.6 A, respectively. Besides, the transient SCC amplitudes in conventional and modular arrangements are 117.8 A

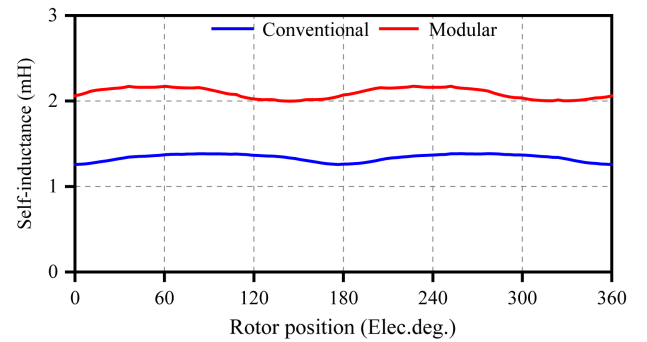


Fig. 8. (Color online) Self-inductance waveforms of two winding arrangements.

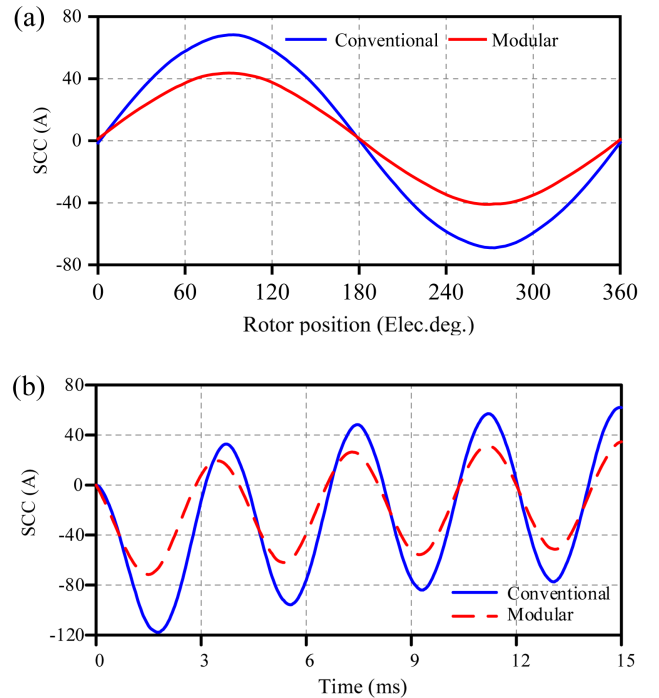


Fig. 9. (Color online) SCC comparison of different winding arrangements: (a) Steady, (b) Transient.

and 71.6 A, respectively. The steady and transient SCC amplitudes are reduced by 36.2 % and 39.2 %, respectively, which confirms that the modular arrangement is beneficial to improving the safe operation in PM machines for flywheel battery.

4.3. Mutual Inductance

The magnetic isolation capability is very significant for fault-isolation in PM machines for flywheel battery, and it

can be embodied through mutual inductance. Fig. 10 compares the mutual inductances of various winding arrangements. Results show that the M_{AB} and M_{AC} in modular one is almost reduced to zero, which indicates that there is no magnetic coupling between phases. That is to say, the modular arrangement has better fault-isolation capability than that of conventional one. In addition, the armature magnetic lines distribution with

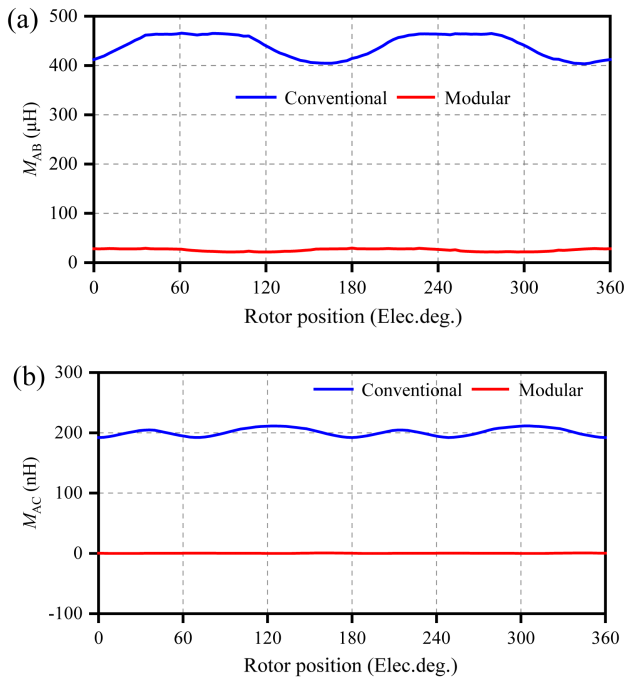


Fig. 10. (Color online) Mutual inductance comparison of different winding arrangements: (a) M_{AB} , (b) M_{AC} .

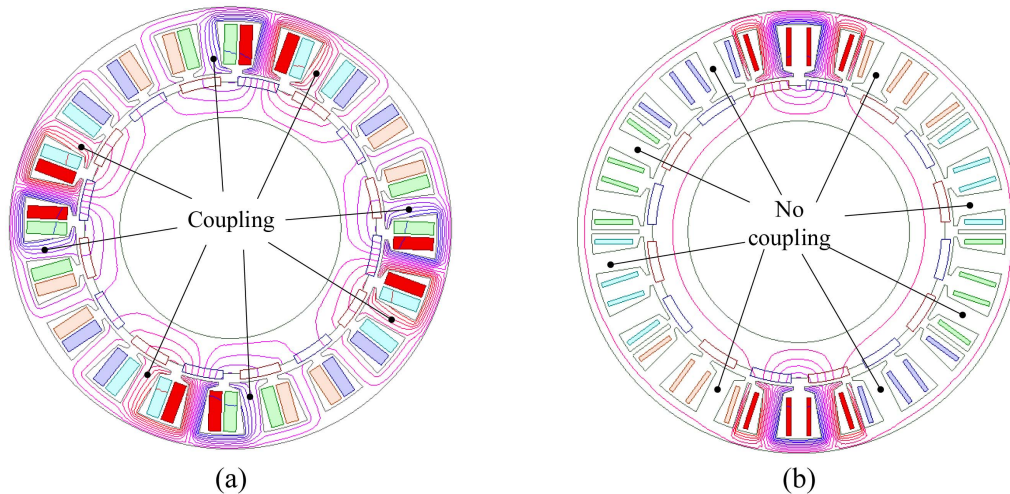


Fig. 11. (Color online) The armature magnetic lines distribution with supplied current in phase-A: (a) Conventional arrangement, (b) Modular arrangement.

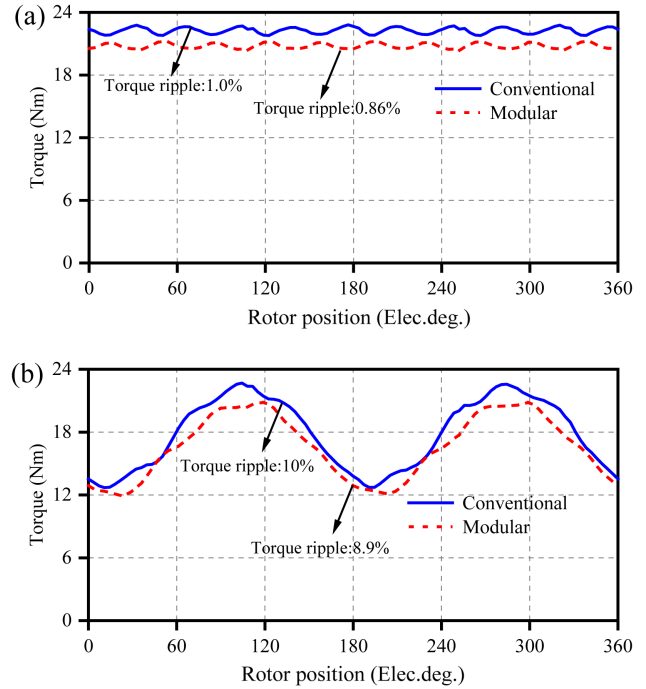


Fig. 12. (Color online) Torque performance comparison under different operation situation: (a) Healthy operation, (b) Phase-A open circuit operation.

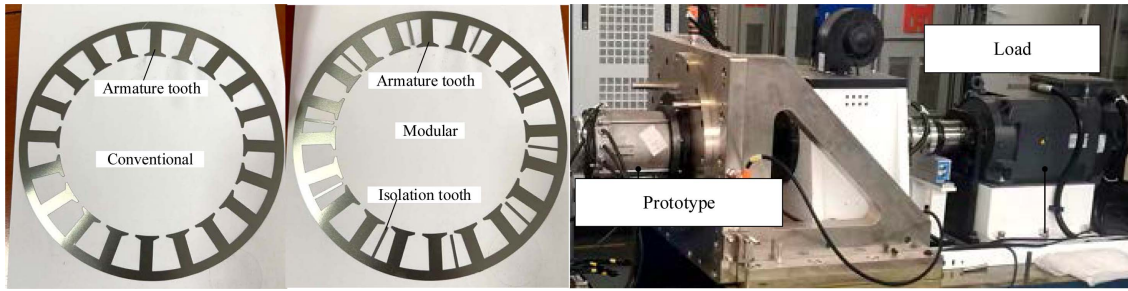


Fig. 13. (Color online) The experimental platform.

supplied current in phase-A is shown in Fig. 11. It can be seen that the armature magnetic lines only distribute in phase-A, while the magnetic lines in the conventional one distribute throughout the whole circumference of the stator. It also verifies that the modular winding arrangement has the advantage of suppressing the magnetic coupling.

4.4. Torque performance

The torque waveforms of conventional and modular winding arrangements under healthy operation as well as phase-A open circuit operation are shown in Fig. 12. Regardless of the operation situation, it can be seen that the average torque in modular arrangement is slightly lower than that in conventional one, which is due to the fundamental amplitude of modular arrangement is reduced. However, the modular arrangement is beneficial for torque ripple suppression.

5. Experimental Validation

Two prototypes with conventional and modular winding arrangement are manufactured to verify the theoretical analysis. The prototypes and stator laminations are shown in Fig. 13. The self-inductance waveforms through FEM and experiment in conventional and modular arrangements are shown in Fig. 14. The experimental values are slightly lower than those simulated by FEM, which is due to the end winding effect and experimental errors. Taking phase-A as an example, the short-circuit fault occurs in phase-A is performed through connecting the outlet terminals of the A+ and A-, while the rest phases are disconnected. The short-circuit current waveform is measured through current clamp and oscilloscope. Fig. 15 compares the FEM simulated and experimental SCC waveforms of the two winding arrangements. It can be seen that the simulated values are consistent with the experimental ones. As a whole, the experimental results indicate that the prototypes are well processed and the validation of

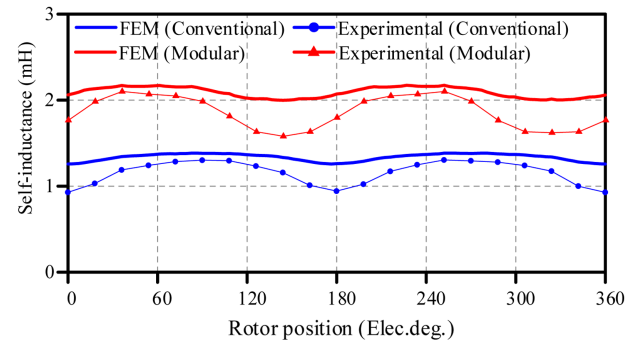


Fig. 14. (Color online) Self-inductance waveforms of two winding arrangements obtained from FEM and experiment.

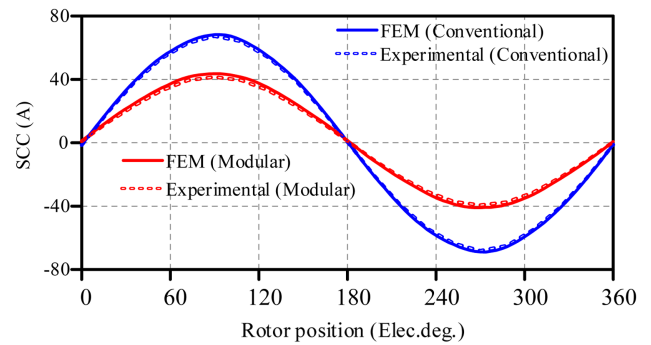


Fig. 15. (Color online) SCC waveforms of two winding arrangements obtained from FEM and experiment.

theoretical analysis.

6. Conclusion

In this paper, a modular winding arrangement has been proposed to improve the fault-tolerant capability of PM machines for flywheel battery and its design principle is elaborated. The distribution characteristics of MMFs in various arrangements are obtained through winding function. The MMF harmonic orders in conventional arrangements satisfies $4(5k \pm 2)$, while that in modular arrangement meets $2(5k \pm 1)$. The inversely proportional

relationship between SCC amplitude and self-inductance is derived through the equivalent circuit diagram in case of phase A short-circuit. The average values of self-inductance in conventional and modular arrangement are 1.34 mH and 2.1 mH, respectively. The mutual inductances in modular arrangement are almost 0, which indicates this arrangement has a better fault-isolation capability than its counterpart one. In addition, the SCC amplitude is reduced by 36.2 % in modular arrangement, which can be attributed to its higher self-inductance. The corresponding experiments are carried out for validation. In conclusion, the modular arrangement is beneficial to suppressing the SCC amplitude and improving fault-isolation capability, which is important for the safety operation of PM machines for flywheel battery.

Acknowledgments

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